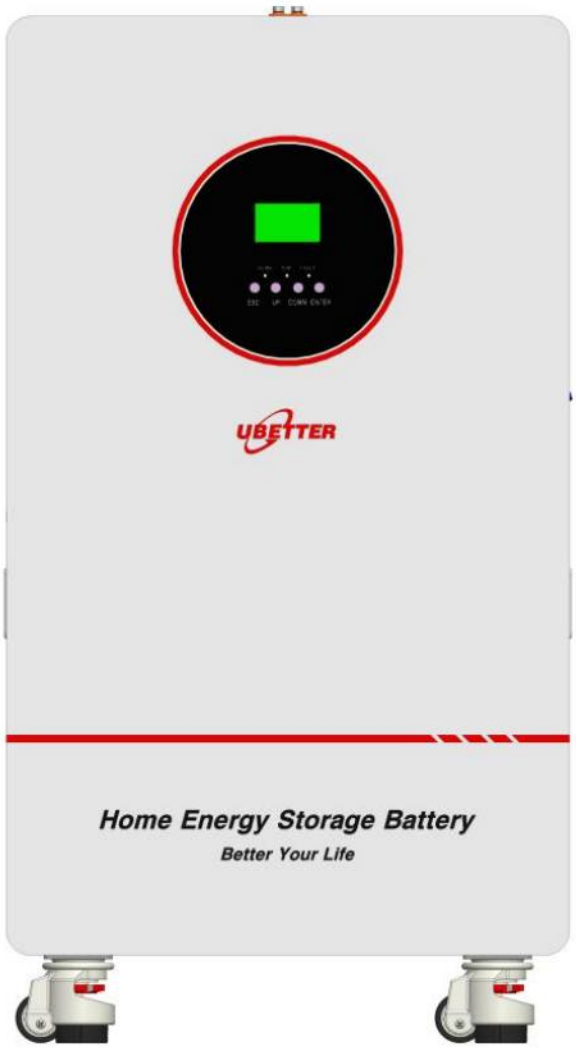


UBETTER TECHNOLOGY Co., Ltd

Home Storage 8KWH+3KW Inverter

All-in-One Machine

product specification sheets

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1.explain

This specification applies to 25.6V 314AH 8S1 P 3.2V 314AH lithium iron phosphate (LiFePO4)rechargeable battery pack manufactured by UBETTER Technology Co.,Ltd.

2.main specifications

2.1 Battery Cell Specifications TED

	No.	project	general parameters	remark
Cell	1	nominal capacity	314Ah	
	2	nominal voltage	3.2V	average operating voltage
	3	internal impedance	0.17±0.05 mΩ	AC 1 KHz Test(50%SOC)
	4	size	(71.7±0.5)*(173.9±0.8) *(207.2±0.8)	Initial size
	5	weight	5.60±0.15kg	about
	6	charging cut-off voltage	3.65 V	Cell temperature T>0℃Cell temperature T>0℃
	7	discharge cut-off voltage	2.5 V	Battery temperature T>0℃ Cell temperature T>0℃
			/	/
	8	charging rate	Standard magnification:0.5P	25℃
	9	discharge rate	Standard magnification:0.5P	25℃
			Maximum continuous discharge rate:1P	25℃
	10	storage temperature	Temperature:-10℃~35℃,humidity ≤85%	
	11	cycle performance	25±2℃0.5C/0.5C 6000 times	Capacity retention≥70%

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2.2.Battery pack specifications

	No.	project	general parameters	remark
battery pack	1	compound mode	8S1P	
	2	rated capacity	314Ah	Standard discharge after standard charging(packaging)
	3	nominal voltage	25.6V	average operating voltage
	4	discharge cut-off voltage	20V	discharge cut-off voltage
	5	charging cut-off voltage	29.2V	charging cut-off voltage
	6	internal impedance	≤50mΩ	Internalresistance measured at 1 kHz AC,measured after 50%charge must benew batteries with less than 5 cycles within one week of shipment.
	7	standard charging	Constant current 150A constant voltage see no.5 0.02CA cutoff	Charging time:about 2 hours
	8	Maximum continuous chargingcurrent	150A	
	9	standard discharge	Constant current:150A	
	10	maximum continuous dischargecurrent	150A	
	11	operating temperature range	Charge:0~60℃	60±25%RH
			Discharge:-30° C~60° C	
	12	storage temperature range	within 12 months	60±25%R.H.
			within 3 months	
			within 7 days	
	13	size	820*450*230.5mm	
	14	weight	Approx: 70 kg	

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3.Battery Management System Specifications

3.1.BMS function introduction

(1) The BMS is designed for 16-string lithium batteries

(2) BMS has all of the following functions:

- Overcharge detection function
- Overdischarge detection function
- Overcurrent detection function
- Short circuit detection function
- temperature detection function
- balance function
- communication function
- alarm function
- total capacity function
- Store history function

3.2.BMS Protection Parameters

project	detail	standard
monomer overcharge protection	overcharge detection voltage	3.7±0.025V
	Overcharge detection delay time	Typical:1.0s
	overcharge release voltage	3.38±0.05V
monomer overdischarge protection	overdischarge detection voltage	2.7±0.05V
	Overdischarge detection delay time	Typical:1.0s
	overdischarge release voltage	2.95.0±0.05V
overcurrent protection	Discharge overcurrent protection current 1	220
	Discharge overcurrent detection delay time 1	1S
	Discharge overcurrent protection current 2	≥300A
	Discharge overcurrent detection delay time 2	100ms
	charging overcurrent protection current	220±10A
short-circuit protection	short circuit protection function	yes
	protection conditions	load short circuit
	load is removed	automatic release

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	protection release condition	charging release
temperature protection	Charging high temperature protection	65℃
	Charge High Temperature Recovery	55℃
	discharge high temperature protection	70℃
	discharge high temperature recovery	60℃
	charging low temperature protection	-5℃
	charging cold recovery	0℃
	discharge cryogenic protection	-20℃
	discharge low temperature recovery	-15℃
balance	equilibrium threshold voltage	3.50V
communication	With RS232 standard communication interface, it can monitor the battery pack capacity, voltage and current environmental temperature in real time. degree and charge/discharge current	
alarm	With over-temperature, over-charge, under-voltage, over-current, short-circuit alarm function	

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4.appearance and structure

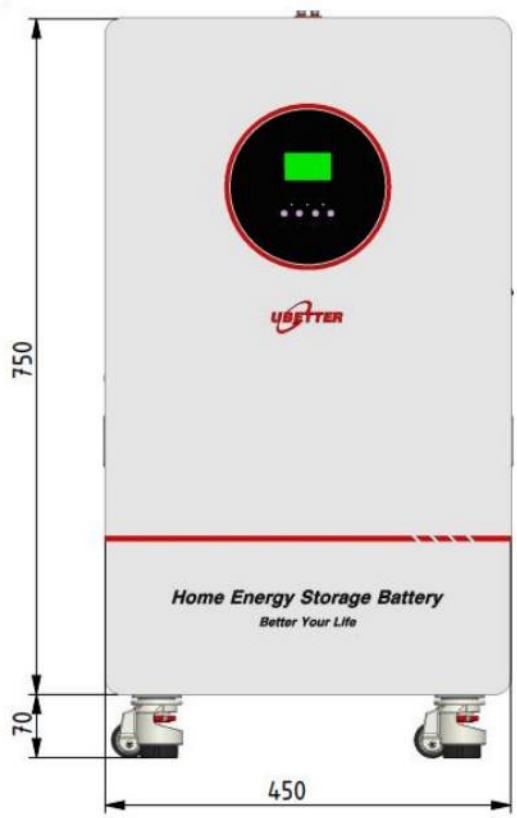
There shall be no scratches,burrs and other mechanical scratches,and the connector shall be free of rust and dirt.For structure and size,please refer to drawings attached to the battery.



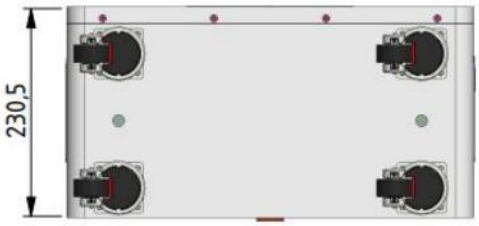
L(length)	820mm	W(width)	450mm	H(thickness)	230.5mm
weight	70kg	the largest charge amount	150A	maximum discharge	150A
Unit(mm)					

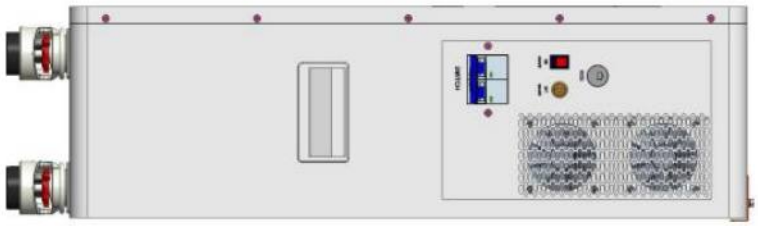
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4.1Product size drawing



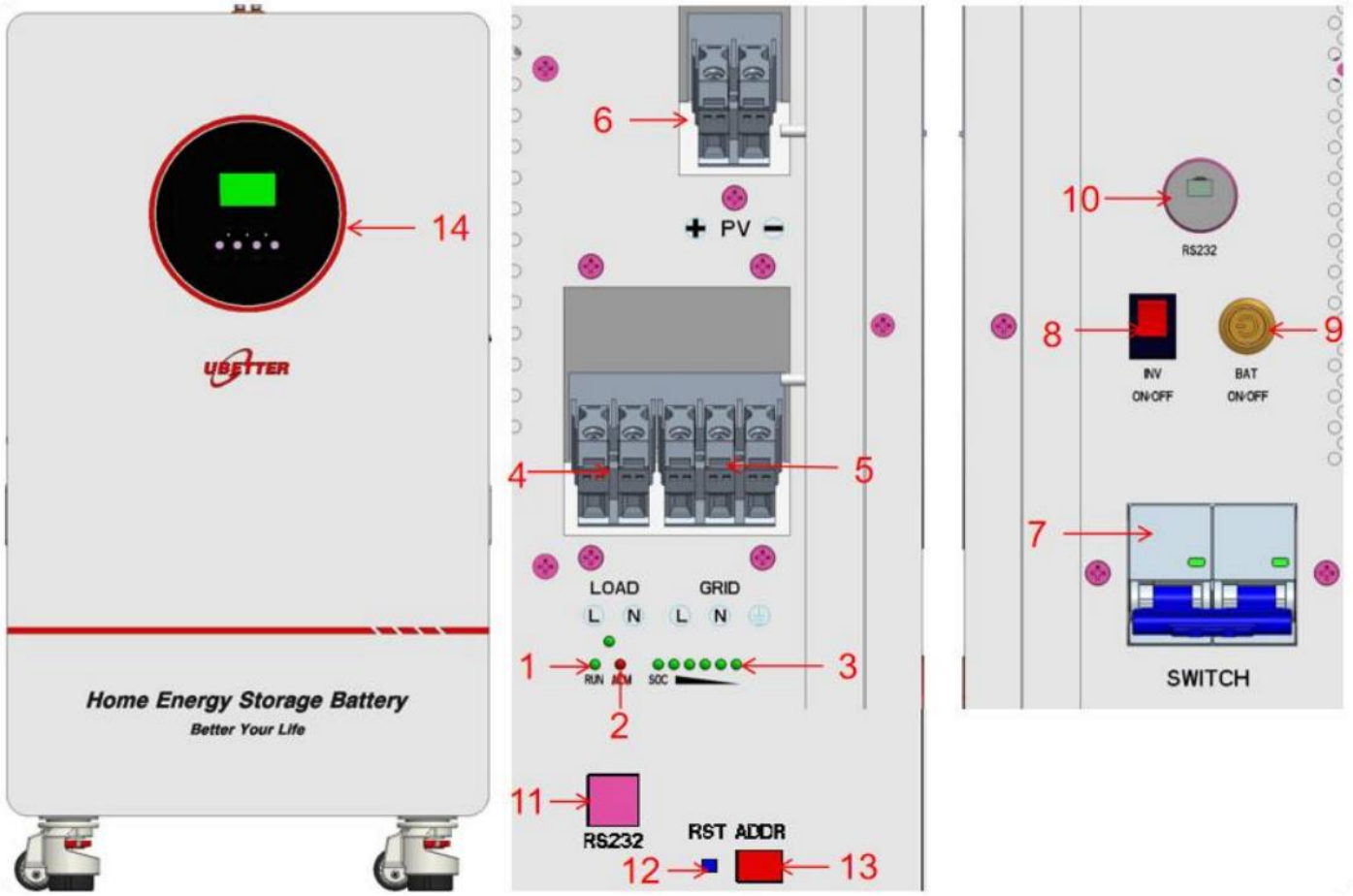






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4.2Battery pack housing structure Pack



Function and interface definition

No.	describe	silk-screen	remark
1	Running indicator light	RUN	Running status indicator
2	alert indicator	ALM	fault warning lamp
3	power display	SOC	Help users intuitively understand the remaining battery
4	amphenol connector	LOAD	load line
5	amphenol connector	GRID	Connected to mains
6	amphenol connector	PV	Connect photovoltaic line
7	air switch	SWITCH	short-circuiter
8	inverter switch	INV ON/OFF	Start/close inverter power supply
9	low current switch	BAT ON/OFF	Turn on/off battery power
10	WIFI terminals	RS232	

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11	upper computer	RS232	upper computer communication
12	reset button	RST	
13	address communication	ADDR	combined dialing address
14	screen	/	/

5. battery test equipment

5.1 size measuring instrument

size measurements shall be made using instruments with an accuracy of not less than 0.1 mm.

5.2 voltmeter

The standard level or more sensitive category specified in the national standard should be used,and the internal resistance should not be less than 10 kiloohms/volt.

5.3 ammeter

Standard levels or more sensitive categories specified in national standards should be used.Total external resistance including ammeter and lead shall be less than 0.01 ohms.

5.4 impedance meter

Impedance shall be measured using the sinusoidal alternating current method(AC 1kHz LCR meter).

6. standard test conditions

The test shall be conducted within one month after the batteries are shipped from our factory and the number of battery cycles before testing shall not exceed five.Unless otherwise defined,tests and measurements shall be carried out at a temperature of 23±2°C,a relative humidity of less than 75%,and an atmospheric pressure of 86kPa to 106kPa.Unless otherwise defined,there shall be a 30-minute rest period after charging and a 30-minute rest period after discharging.

7. storage and other

7.1 long-term storage

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If stored for a long time(unused,more than three months),batteries should be stored in a dry and cool place.The storage voltage of the battery shall be 48V-50V and the battery shall be stored at a temperature of 23±2℃and a humidity of 45%-75%.Batteries that have not been used for a long time should be recharged every three months to ensure that the battery voltage is within the above range.

7.2 else

Any matters not covered by this specification shall be settled by consultation between Customer and UBETTER.

8. Revision of this specification

This specification is subject to change upon prior notice from UBETTER.

9. appendix

Lithium-ion rechargeable battery handling precautions and guidelines

10. introduction

This document,Precautions and Guidelines for Handling Lithium Ion Rechargeable Batteries,applies to battery cells manufactured by UBETTER.

Note(1):If the customer requires other applications or operating conditions not described in this document,the customer should contact UBETTER in advance.Additional experiments may be required to verify performance and safety under these conditions.

Note(2):UBETTER will not be responsible for any accident if the battery is used outside the conditions described in this document.

Note(3):If necessary,UBETTER will inform the customer in writing about improvements to the correct use and handling of batteries.

Danger warning!

-Do not immerse the battery in water or damp it.

-Do not use or store batteries near sources of heat such as fire or heaters.

-Do not use any chargers not recommended by UBETTER.

-Do not reverse the positive(+)and negative(-)terminals.

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-Do not connect the battery directly to a wall socket or car cigarette lighter socket.

-Do not put batteries into fire or heat them directly.

-Do not short-circuit the battery by connecting wires or other metallic objects to the positive(+)and negative(-)terminals.

-Do not puncture the battery case with nails or other sharp objects,and do not crack or trample on it with a hammer.

-Do not knock,throw or apply violent physical impact to the battery.

-Do not solder battery terminals directly.

-Do not disassemble or modify the battery in any way.

-Do not put batteries in microwave ovens or pressurized containers.

-Do not combine batteries with primary batteries(e.g.dry cells)or batteries of different capacities,types or brands.

-Do not use batteries if they smell,generate heat,discolor or deform,or look abnormal in any way.If the battery is in use or charging,remove it from the device or charger immediately and discontinue use.

Warning!

Do not use or store batteries in places exposed to extreme temperatures,such as direct sunlight under car windows in hot weather.Otherwise,the battery may overheat.This also reduces battery performance and/or service life.If the battery leaks and electrolyte gets into your eyes,do not rub it.Instead,rinse with clean,running water and seek immediate medical help.Electrolytes can cause eye injury if left unattended.

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